

WILSY OS Knowledge Base Production Playbook - FG109

User guide, investor guide, and engineering release guide for the production Knowledge Base

Document class: Investor, user, and engineering knowledge base
Prepared for: Investors, users, operators, auditors, and future engineers
Generated by: Wilson Khanyezi
Source posture: KNOWLEDGE_BASE_VERIFIED
PDF route: /api/generate/pdf

KNOWLEDGE BASE PURPOSE

Wilsy OS Knowledge Base is the production memory layer for the platform. It turns completed component work, stable releases, evidence routes, operating rules, and user workflows into searchable, printable, permissioned knowledge.

The Knowledge Base must not become a loose folder of PDFs. Each important update must ship with a manifest entry, source markdown, machine-readable JSON, saved PDF, proof sidecar, searchable metadata, and a release guard path.

- Users get plain-language operating guidance.
- Investors get evidence that the product can explain, prove, and govern itself.
- Engineers get release contracts that prevent future regressions.
- Auditors get proof that the PDF, JSON, source posture, owner, and hashes align.

EXECUTIVE SUMMARY

The Knowledge Base is now ready to behave like a production operating surface: it has a consolidated operating bar, live search, manifest-backed saved artifacts, proof records, permission-aware Vault routes, and controlled document actions.

FG109 adds the missing production ritual. Every completed component or UI surface should now produce a user guide, investor guide, engineering guide, JSON companion, PDF export, and proof sidecar before the work is treated as release complete.

- Production posture: saved artifact access, proof sidecar verification, and permissioned routes.
- Commercial posture: each feature can be explained to buyers, operators, and investors from the product itself.
- Engineering posture: every future component receives a documented operating contract and test/guard checklist.

USER OPERATING GUIDE

Users should open the Knowledge Base when they need a trusted answer about how a Wilsy OS surface works. The Vault should answer from saved artifacts only, then let the user open the PDF, inspect evidence, download, print, copy fingerprints, or open the companion JSON.

- Search by title, feature, audience, owner, proof posture, release label, category, or fingerprint.

- Use Open to preview the saved PDF inside the workspace.
- Use JSON to inspect the machine-readable playbook for training, automation, and AI retrieval.
- Use Evidence to inspect the proof sidecar and hash alignment.
- Use Copy fingerprint when a customer, auditor, or support case needs immutable proof.

INVESTOR READINESS GUIDE

The investor story is not that Wilsy OS has documentation. The investor story is that Wilsy OS turns product execution into governed institutional memory. A feature does not disappear into code; it becomes an artifact with purpose, proof, user training, and engineering continuity.

This creates a scalable diligence surface. Investors can inspect whether the platform knows what was shipped, why it matters, how it is used, how it is controlled, and what evidence proves it.

- Every strategic component can become a diligence-ready Knowledge Base asset.
- The PDF is the boardroom-readable artifact.
- The JSON is the machine-readable artifact for search, AI retrieval, and future automation.
- The proof sidecar binds the PDF to SHA3, owner identity, source posture, required text, and forbidden text.
- The manifest makes the document discoverable in the authenticated Vault.

ENGINEERING RELEASE GUIDE

When a component or UI surface is created or materially updated, the release is not complete until the Knowledge Base artifact is created or updated. This makes production knowledge part of the engineering definition of done.

- Inspect JSX ownership, backend routes, guards, state, effects, and evidence contracts before writing docs.
- Create or update the markdown playbook in docs/knowledge-base.
- Create or update the JSON companion with sections, audiences, routes, source files, release checklist, and evidence sources.
- Generate or refresh the PDF through the approved artifact workflow.
- Generate or refresh the proof sidecar with PDF SHA3 and canonical section declarations.
- Add or update the manifest entry with pdfPath, jsonPath, proofPath, audiences, permissionScope, search terms, requiredPdfText, and forbiddenPdfText.
- Run documentation guard, secret guard, Knowledge Base guard, and the client build when the surface touches frontend code.

RUNTIME CONTRACT

The Knowledge Base runtime contract is source discipline. The Vault reads saved artifacts from the manifest. It should not regenerate documents just because a user wants to read, print, download, or inspect evidence.

- Manifest source: client/src/data/wilsyKnowledgeBaseManifest.js
- Vault list route: /api/knowledge-base/vault
- Saved PDF route: /api/knowledge-base/vault/:id/pdf
- Saved JSON route: /api/knowledge-base/vault/:id/json
- Proof route: /api/knowledge-base/vault/:id/proof
- Generation route: /api/generate/pdf
- Forbidden legacy route: /api/crm/leads/views/proof-pack/export/pdf

PDF AND EVIDENCE ROUTE CONTRACT

The PDF and JSON artifacts are saved Knowledge Base assets. The proof sidecar is the verification record. These three files serve different jobs and should not be collapsed into one generic blob.

- PDF: human-readable guide for users, investors, operators, and auditors.
- JSON: structured guide for Knowledge Base search, AI retrieval, training flows, and automation.
- Proof JSON: hash-bound verification record for release integrity.
- All saved artifact routes must preserve authentication, institutional request headers, and no-regeneration discipline.

FORENSIC INTERNET EVIDENCE

This playbook maps Wilsy OS Knowledge Base production behavior to primary external evidence from NIST, OWASP, W3C, ISO, and the South African Government. The point is not decorative citation. The point is to make Wilsy OS release discipline auditable against recognized risk, accessibility, security, AI governance, and privacy controls.

- NIST AI RMF supports trustworthy AI risk management.
- NIST SSDF supports secure software release discipline.
- OWASP LLM Top 10 2025 supports AI operator threat modeling.
- W3C WCAG 2.2 supports accessible Knowledge Base operation.
- ISO/IEC 42001 supports AI management system governance.
- POPIA supports South African privacy and automated decision-making obligations.

KNOWLEDGE BASE AUTHORITY AND RETENTION

Authority belongs to the manifest, the saved files, the proof sidecar, and the release guard. The Knowledge Base should preserve institutional confidence by keeping user-facing text professional and by hiding raw backend implementation details unless an operator deliberately opens evidence.

- Generated By must use a professional display name: Wilson Khanyezi.
- Source posture must remain KNOWLEDGE_BASE_VERIFIED for locked production assets.
- Forbidden legal-template placeholders must never appear in Knowledge Base PDFs.
- Every new component playbook must remain searchable by users, investors, and future engineers.

RELEASE CADENCE

Trigger: Every completed component update, new UI component, stable release, or production rescue that changes user workflow or evidence posture.

- Required artifact: Markdown playbook
- Required artifact: Machine-readable JSON companion
- Required artifact: Saved PDF
- Required artifact: Proof sidecar JSON
- Required artifact: Manifest entry
- Required artifact: Guard evidence
- Gate: node scripts/wilsy-documentation-guard.js

- Gate: node scripts/wilsy-secret-guard.js
- Gate: node scripts/wilsy-knowledge-base-guard.js --locked
- Gate: cd client && npm run build

FORENSIC INTERNET EVIDENCE SOURCES

NIST AI Risk Management Framework

<https://www.nist.gov/itl/ai-risk-management-framework>

Use Govern, Map, Measure, and Manage discipline as the control language for AI-enabled operating knowledge.

NIST SP 800-218 Secure Software Development Framework

<https://csrc.nist.gov/pubs/sp/800/218/final>

Require secure development practices, vulnerability reduction, and supplier communication before production release.

OWASP Top 10 for LLM and GenAI Applications 2025

<https://genai.owasp.org/llm-top-10/>

Treat prompt injection, sensitive information disclosure, supply chain, excessive agency, misinformation, and unbounded consumption as AI operator release gates.

W3C Web Content Accessibility Guidelines 2.2

<https://www.w3.org/TR/WCAG22/>

Keep Knowledge Base search, buttons, PDF preview, and document actions perceivable, operable, understandable, and robust.

ISO/IEC 42001:2023 AI Management System

<https://www.iso.org/standard/42001>

Frame Wilsy AI and Knowledge Base governance as a maintained management system with policies, controls, measurement, and improvement.

Protection of Personal Information Act 4 of 2013

<https://www.gov.za/documents/protection-personal-information-act>

Treat privacy, lawful processing, cross-border information flow, data subject rights, and automated decision-making as Knowledge Base training obligations.

FILE CONTRACT

- Markdown path: docs/knowledge-base/WILSY-OS-Knowledge-Base-Production-Playbook-FG109.md
- JSON path: docs/knowledge-base/WILSY-OS-Knowledge-Base-Production-Playbook-FG109.json
- PDF path: docs/knowledge-base/WILSY-OS-Knowledge-Base-Production-Playbook-FG109.pdf
- Proof path: docs/knowledge-base/WILSY-OS-Knowledge-Base-Production-Playbook-FG109.proof.json
- Vault route: /api/knowledge-base/vault
- Generation route: /api/generate/pdf
- Forbidden route: /api/crm/leads/views/proof-pack/export/pdf